

Contribution of dietary protein to sulfide production in the large intestine: an in vitro and a controlled feeding study in humans.

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BACKGROUND: Hydrogen sulfide is a luminally acting, bacterially derived cell poison that has been implicated in ulcerative colitis. Sulfide generation in the colon is probably driven by dietary components such as sulfur-containing amino acids (SAAs) and inorganic sulfur (eg, sulfite). **OBJECTIVE:** We assessed the contribution of SAAs from meat to sulfide production by intestinal bacteria with use of both a model culture system in vitro and an in vivo human feeding study. **DESIGN:** Five healthy men were housed in a metabolic suite and fed a sequence of 5 diets for 10 d each. Meat intake ranged from 0 g/d with a vegetarian diet to 600 g/d with a high-meat diet. Fecal sulfide and urinary sulfate were measured in samples collected on days 9 and 10 of each diet period. Additionally, 5 or 10 g bovine serum albumin or casein/L was added to batch cultures inoculated with feces from 4 healthy volunteers. Concentrations of sulfide, ammonia, and Lowry-reactive substances were measured over 48 h. **RESULTS:** Mean ($\pm$ SEM) fecal sulfide concentrations ranged from 0.22 $\pm$ 0.02 mmol/kg with the 0-g/d diet to 3.38 $\pm$ 0.31 mmol/kg with the 600-g/d diet and were significantly related to meat intake (P : < 0.001). Sulfide formation in fecal batch cultures supplemented with both bovine serum albumin and casein correlated with protein digestion, as measured by the disappearance of Lowry-reactive substances and the appearance of ammonia. **CONCLUSION:** Dietary protein from meat is an important substrate for sulfide generation by bacteria in the human large intestine.